

Smart School Management System

Enterprise Architecture, Flowcharts, ER Diagram
& 45-Module Technical Specification

COMPANY	Infosof Technologies
PREPARED BY	Rupam Saha (Technical Support)
SUBMITTED TO	Monisha Sirkar (Manager)
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Project Snapshot

This report documents the architecture, functional modules, data model, workflows, security posture and verification results of the **Smart School Management System**, prepared by **Rupam Saha** (Technical Support), **Infosof Technologies**, for review by **Monisha Sirkar**, Manager.

Item	Detail
Project	Smart School – School Management System (ERP)
Company	Infosof Technologies
Prepared by	Rupam Saha, Technical Support
Submitted to	Monisha Sirkar (Manager)
Document version	2026.1.0
Backend	CodeIgniter 3.x / PHP 8.x (tested with PHP 8.3)
Frontend	HTML5 / CSS3 / Vanilla JavaScript Single Page Application (zero build tools)
Database	MySQL 8.x (InnoDB, utf8mb4_unicode_ci, 56 relational tables)
Payments	Razorpay Standard Checkout with HMAC-SHA256 verification
Hosting	cPanel shared/VPS, Apache / LiteSpeed on Linux or Windows

45	8	56	45/45
FUNCTIONAL MODULES	USER ROLES	DATABASE TABLES	TESTS PASSED

Document Version: 2026.1.0

Project: Smart School – School Management System

Client / Specification: Infosof Technologies 2026

Backend Framework: CodeIgniter 3.x / PHP 8.x (Tested with PHP 8.3 & MySQL 8.4)

Frontend Architecture: Modern HTML5 / CSS3 / Vanilla JavaScript Single Page Application (Zero Build Tools)

Database: MySQL 8.x / 5.x (InnoDB, utf8mb4_unicode_ci, 56 Relational Tables)

1. Executive Summary & System Overview

The **Smart School Management System** is an enterprise-grade school enterprise resource planning (ERP) platform designed to automate and streamline academic, administrative, financial, logistical, and communication operations across educational institutions.

1.1 Core Value Proposition

- **Full Operational Coverage:** Implements 45 discrete functional modules covering the entire student and staff lifecycle from initial admission inquiries to graduation and alumni tracking.
- **Lightweight, Zero-Build Frontend:** Built with browser-native JavaScript (ES6+), semantic HTML5, and responsive Vanilla CSS. It requires no Node.js compilation, npm bundling, or webpack/vite build steps, ensuring instant load times and effortless maintenance.
- **Enterprise PHP 8.x / CodeIgniter Backend:** High-speed RESTful JSON APIs running on CodeIgniter 3.x modernized for PHP 8.3, utilizing strict typing, PDO database abstractions, and token-based stateless security.
- **Multi-Tenant & Multi-Campus Ready:** Native support for multi-branch campus operations, dynamic session switching, and institutional customization.
- **Real-Time Multi-Tab Synchronization:** Integrated browser `BroadcastChannel` and `storage` event synchronization engine allowing changes made in one browser tab (such as attendance marking or fee payments) to reflect instantaneously across all open windows without manual refresh.

1.2 Role-Based Access Control (RBAC) Matrix

The system enforces strict permission checks across 8 distinct user roles:

Role Name	Description & Primary Responsibilities	Scope of Access
Super Admin	System owner with unrestricted rights across all institutions and configuration.	Global (All 45 Modules, Database, Settings, 2FA, Multi-Campus)
Admin	School principal / senior administrative officer managing day-to-day operations.	School-wide (Academics, Staff, Finance, Students, Reports)
Teacher	Faculty members handling classroom instruction, homework, attendance, and grading.	Classroom Scope (Class Timetable, Attendance, Marks Entry, Live Classes)
Accountant	Financial officer managing fee collection, receipts, expenses, and ledgers.	Financial Scope (Fees Collection, Quick Fees, Expense Head, Thermal POS)
Librarian	Information officer managing book inventory, borrowing, and returns.	Library Operations (Catalog, Book Issues, Returns, Overdue Fines)
Receptionist	Front-desk staff handling visitor inquiries, student admissions, and phone calls.	Front Office (Admissions, Inquiries, Notices, Calendar)
Student	Enrolled learner viewing study materials, schedules, fee dues, and reports.	Self-Profile (View Timetable, Grades, Fees Due, Pay Online, Download Materials)
Parent	Guardian monitoring academic progress, attendance, and paying tuition fees.	Ward-Profile (Track Children, Attendance Alerts, Online Fee Clearance)

2. System Architecture & Technical Stack

2.1 High-Level Architecture Diagram

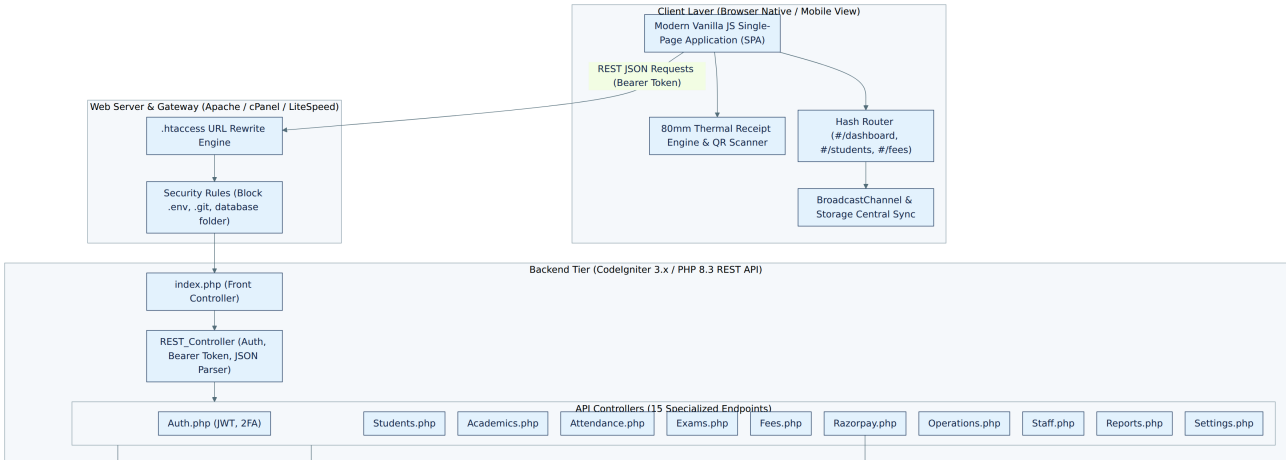


Figure 1: High-level system architecture (part 1 of 2)

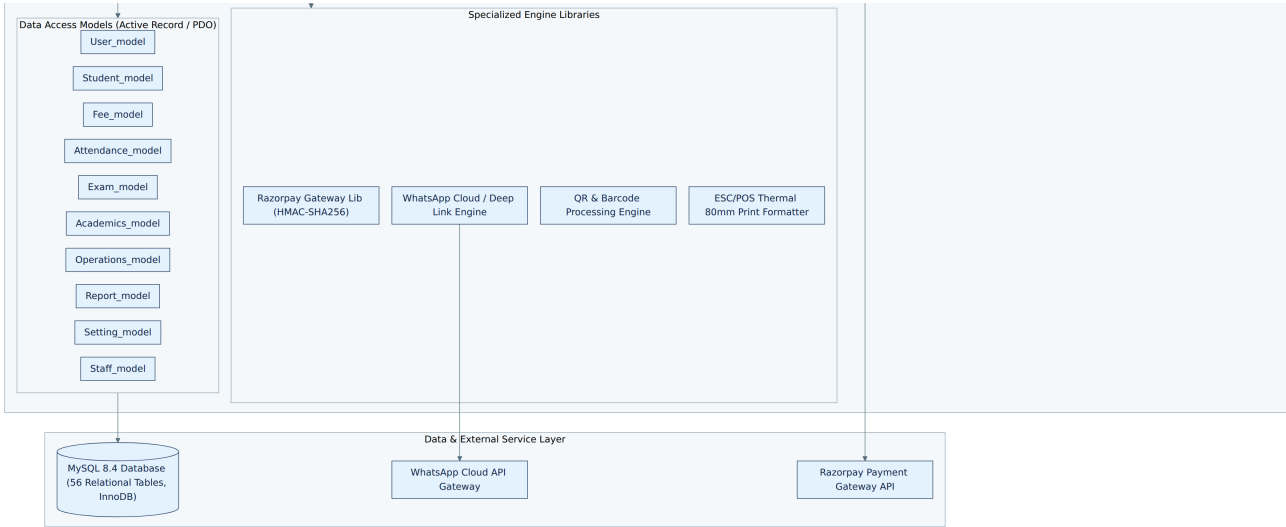


Figure 1: High-level system architecture (part 2 of 2)

2.2 Technical Stack Specifications

- **Operating Environment:** Windows, Linux, Ubuntu, cPanel (Shared & VPS Hosting).
- **Backend Framework:** CodeIgniter 3.x (PHP 8.3 compatible), strictly configured for RESTful JSON responses.
- **Frontend Stack:** HTML5, CSS3, ES6+ JavaScript, CSS Grid & Flexbox, SVG vector icons.
- **External Dependencies:** Zero npm dependencies, zero Node runtime dependencies for the frontend.
- **Payment Processing:** Official Razorpay Standard Checkout & Orders API with HMAC-SHA256 signature verification and simulated test sandbox.
- **Hardware Integration:** Standard 80mm ESC/POS thermal receipt printers (via window.print CSS styling) and Camera-based QR/Barcode scanning.

3. Complete 45-Module Functional Specification

Below is the exhaustive, feature-by-feature breakdown of all 45 modules certified under the Infosof Technologies 2026 specification:

#	Module	#	Module
1	User & Role Management	24	Transfer Certificate (TC) Generator
2	Admin Telemetry Dashboard	25	Admit Card & Hall Ticket Generator
3	Student Admission Management	26	Annual Academic Calendar & Events
4	360° Comprehensive Student	27	Dynamic Custom Fields Builder
5	Multi-Criteria Student Search	28	QR Code / Barcode Attendance Scanner
6	Student Session Promotion	29	Two-Factor Authentication (2FA TOTP)
7	Caste & Category Allocation	30	Behavior Records & Merit/Demerit Logging
8	Fees Management & Invoicing	31	Thermal 80mm POS Receipt Printing
9	Income & Expense Ledgers	32	Quick Fee Creation & Counter Cash Entry
10	Daily Attendance Roll Call	33	Automatic Overdue Fine Calculation Engine
11	Examination & CBSE 9-Point	34	Fee Concession & Discount Management
12	Class & Section Management	35	Online Payment Processing (Razorpay Gateway)
13		36	
14	Download Center (E-Learning)	37	Multi-School & Multi-Campus Switching
15		38	
16	Transport & Vehicle Tracking	39	Front Website Public Portal & WhatsApp Chat Widget
17	Hostel Wing & Room Allotment	40	Academic Session Transition & Rollover
18	Notice Board & Announcements	41	Student Digital Document Vault
19		42	
20	Live Classes (Zoom / Meet)	43	RTE (Right to Education) Quota Management
21		44	
22		45	
23			

Module 01 User & Role Management	
Description	Centralized identity and privilege governance system.
Capabilities	Supports 8 standardized user roles (<code>super_admin</code> , <code>admin</code> , <code>teacher</code> , <code>accountant</code> , <code>librarian</code> , <code>receptionist</code> , <code>student</code> , <code>parent</code>). Supports role assignment, granular module permissions, account disabling, and password reset.
API endpoints	GET <code>/api/users</code> , POST <code>/api/users</code> , PUT <code>/api/users/:id</code> , DELETE <code>/api/users/:id</code> , GET <code>/api/users/roles</code> .
Database tables	<code>users</code> , <code>permissions</code> , <code>role_permissions</code> , <code>sessions</code> .

Module 02 Admin Telemetry Dashboard	
Description	Real-time operational intelligence mission control.
Capabilities	Displays high-priority telemetry cards (Total Students, Staff Count, Fee Collected vs Pending, Attendance Percentages), quick action shortcuts, recent notices, cash flow charts, and pending approvals.
API endpoints	GET /api/dashboard/stats, GET /api/dashboard/charts.
Database tables	Aggregates students, staff, student_fees, attendances, transactions, notices.

Module 03 Student Admission Management	
Description	Formal student enrollment and onboarding pipeline.
Capabilities	Multi-tab registration capture including personal information, Aadhaar/ID, guardian details, address, previous school history, blood group, medical alerts, admission date, and auto-generated Admission Number (ADM-YYYY-XXXX).
API endpoints	POST /api/students, GET /api/students/next-admission-no.
Database tables	students, student_documents, student_custom_field_values.

Module 04 360° Comprehensive Student Profile	
Description	Holistic student dossier providing a single-pane-of-glass overview.
Capabilities	Consolidates attendance rates, fee balances, fee receipts history, exam report cards, behavioral incident history, parent contact details, documents vault, sibling relationships, hostel allocation, and transport stops.
API endpoints	GET /api/students/profile/:id, GET /api/students/:id.
Database tables	students, attendances, student_fees, exam_results, student_notes, student_transports, student_hostels.

Module 05 Multi-Criteria Student Search	
Description	Fast parametric query engine for retrieving student cohorts.
Capabilities	Real-time debounce search matching by name, admission number, roll number, father's mobile number, class, section, gender, and category.
API endpoints	GET /api/students?query=...&class_id=...§ion=...
Database tables	students (indexed on admission_no, first_name, last_name, phone).

Module 06 Student Session Promotion	
Description	Academic year progression and batch rollover engine.
Capabilities	Promotes students from one class/section to the next academic session based on academic standing (Pass / Retain). Preserves historical promotion logs with previous class and section records.
API endpoints	POST /api/academics/promote-students.
Database tables	student_promotions, students, academic_sessions.

Module 07 Caste & Category Allocation	
Description	Institutional demographic categorization and diversity compliance.
Capabilities	Classifies learners into standardized regulatory categories (General, OBC, SC, ST, EWS, Physically Challenged). Enables state scholarship reports and quota verification.
API endpoints	GET /api/students/categories.
Database tables	students.category.

Module 08 Fees Management & Invoicing	
Description	End-to-end fee structure, billing, and accounting suite.
Capabilities	Configures multi-term fee types (Tuition, Laboratory, Annual, Transport, Sports). Generates digital fee invoices, issues unique receipt numbers, tracks partial payments, overdue balances, and historical ledgers.
API endpoints	GET /api/fees/types, POST /api/fees/collect, GET /api/fees/student/:id.
Database tables	fee_types, student_fees, transactions.

Module 09 Income & Expense Ledgers	
Description	General ledger and institutional cash flow management.
Capabilities	Manages double-entry account heads (School Fee, Grants, Canteen, Repairs, Utility Bills, Salaries). Records income and expense vouchers with payment mode, attachments, and date stamps.
API endpoints	GET /api/finance/transactions, POST /api/finance/transactions, GET /api/finance/summary.
Database tables	accounts_heads, transactions.

Module 10 Daily Attendance Roll Call	
Description	Classroom roll-call register and attendance reporting.
Capabilities	Bulk daily marking interface (Present, Absent, Late, Half Day, Holiday). Instant attendance summary calculation, percentage tracking, and automated absence alert generation.
API endpoints	GET /api/attendance/daily, POST /api/attendance/mark, GET /api/attendance/report.
Database tables	attendances, students, school_classes.

Module 11 Examination Management & CBSE 9-Point Grading	
Description	Examination scheduler, marks capture, and grading engine.
Capabilities	Configures terms (Unit Test, Half Yearly, Annual), subjects, maximum and passing marks. Automatically computes percentages, ranks, and assigns official CBSE 9-point letter grades (A1, A2, B1, B2, C1, C2, D, E1, E2).
API endpoints	GET /api/exams, POST /api/exams/marks, GET /api/exams/report-card/:student_id/:exam_id.
Database tables	exams, exam_schedules, exam_results.

Module 12 Academic Class & Section Management	
Description	Institutional academic division configuration.
Capabilities	Configures classes (e.g., Nursery to Class 12), sections (A, B, C, D), assigns dedicated Class Teachers, room numbers, and subject curricula.
API endpoints	GET /api/academics/classes, POST /api/academics/classes, GET /api/academics/subjects.
Database tables	school_classes, subjects, class_subjects.

Module 13 Weekly Class Timetable Matrix	
Description	Multi-period weekly scheduling grid.
Capabilities	Allocates subjects, start/end period times, and assigned faculty across Monday through Saturday. Prevents teacher scheduling clashes.
API endpoints	GET /api/academics/timetable?class_id=...§ion=..., POST /api/academics/timetable.
Database tables	timetables, school_classes, subjects, staff.

Module 14 Download Center (E-Learning Repository)	
Description	Digital educational resource and syllabus depository.
Capabilities	Enables teachers to upload syllabus documents, weekly assignments, past exam question papers, and study materials classified by class and subject with secure file downloads.
API endpoints	GET /api/academics/downloads, POST /api/academics/downloads.
Database tables	download_materials.

Module 15 Library Management & Book Circulation	
Description	Integrated library cataloging and circulation desk.
Capabilities	Tracks book inventory by title, author, ISBN, publisher, rack number, total vs available copies. Handles book issue and return workflows for students and staff with automatic overdue fine calculation.
API endpoints	GET /api/operations/library/books, POST /api/operations/library/issue, POST /api/operations/library/return.
Database tables	library_books, book_issues.

Module 16 Transport Management & Route Stops	
Description	Fleet logistics, vehicle registration, and student route allocation.
Capabilities	Registers school buses and vans, driver names, phone numbers, route itineraries, and designated pickup/drop-off stops with exact timings and monthly transport fees.
API endpoints	GET /api/operations/transport/routes, POST /api/operations/transport/assign.
Database tables	transport_routes, transport_stops, student_transports.

Module 17 Hostel Management & Room Allocation	
Description	Boarding house governance and student room allotment.
Capabilities	Manages hostel buildings (Boys, Girls, Junior, Senior), room types (AC, Non-AC, Single, Shared), intake capacity, current vacancies, and room allotment ledger.
API endpoints	GET /api/operations/hostel/wings, POST /api/operations/hostel/assign.
Database tables	hostels, hostel_rooms, student_hostels.

Module 18 Notice Board & Communications Hub	
Description	Omnichannel circular broadcasting and communications center.
Capabilities	Publishes official notices, alerts, and academic circulars with targeted audience scope (All , Students , Parents , Staff). Role-guarded so students cannot publish circulars.
API endpoints	GET /api/operations/notices, POST /api/operations/notices, DELETE /api/operations/notices/:id.
Database tables	notices.

Module 19 WhatsApp Messaging Integration	
Description	Automated and interactive WhatsApp notifications engine.
Capabilities	Generates instant WhatsApp communication deep links (https://wa.me/...) and backend notification payloads for fee due reminders, absence alerts, examination hall tickets, and circulars.
API endpoints	POST /api/operations/whatsapp/send.
Libraries & tables	Whatsapp_lib.php, notification_logs.

Module 20 Online Classes / Live Classes	
Description	Distance learning and virtual classroom scheduler.
Capabilities	Integrates Zoom and Google Meet session links with meeting IDs, passcodes, scheduled dates, durations, and direct student launch buttons.
API endpoints	GET /api/academics/live-classes, POST /api/academics/live-classes.
Database tables	live_classes.

Module 21 Staff Management & Human Resources	
Description	Comprehensive faculty and employee master directory.
Capabilities	Records employee IDs (EMP-XXXX), designations, departments, academic qualifications, joining dates, emergency contacts, bank account details, and basic salary structures.
API endpoints	GET /api/staff, POST /api/staff, GET /api/staff/:id.
Database tables	staff, salary_records, payslip_items.

Module 22 Staff Attendance & Punch Logs	
Description	Employee attendance and working hour tracking.
Capabilities	Records daily staff presence, punch-in time, punch-out time, half days, and leave applications with approval status.
API endpoints	GET /api/staff/attendance, POST /api/staff/attendance/mark.
Database tables	staff_attendances, leave_applications.

Module 23 Automated Student Curriculum Vitae (CV) Engine	
Description	Auto-generated student academic and co-curricular resume.
Capabilities	Compiles student bio, educational history, extracurricular awards, sports participation, disciplinary record, and attendance percentage into a printable curriculum vitae.
API endpoints	GET /api/students/:id/cv.
Database tables	students, exam_results, student_notes, student_timelines.

Module 24 Transfer Certificate (TC) Generator	
Description	Formal CBSE / State Board compliant Transfer Certificate issuance.
Capabilities	Generates serialized official school leaving certificates (TC-YYYY-XXXX) including student details, date of admission, date of leaving, reason for departure, conduct assessment, and digital signature boxes.
API endpoints	GET /api/students/:id/tc, POST /api/students/:id/issue-tc.
Database tables	students, school_settings.

Module 25 Examination Admit Card (Hall Ticket)	
Description	Examination entry pass and hall ticket generator.
Capabilities	Produces printable hall tickets containing student photo, roll number, class & section, exam schedule matrix (dates, subjects, room numbers), and examination conduct guidelines.
API endpoints	GET /api/exams/admit-card?exam_id=...&student_id=...
Database tables	exams, exam_schedules, students.

Module 26 Annual Academic Calendar & Events	
Description	Centralized institutional event and holiday planner.
Capabilities	Schedules term examinations, sports days, cultural festivals, public holidays, and parent-teacher meetings with visual color coding and dashboard timeline widgets.
API endpoints	GET /api/academics/calendar, POST /api/academics/calendar.
Database tables	calendar_events.

Module 27 Dynamic Custom Fields Builder	
Description	Schema extension engine for custom institutional data capture.
Capabilities	Allows administrators to define dynamic custom form fields (Text, Number, Date, Select Dropdown, Checkbox) attached to the Student Admission or Staff Creation forms without touching SQL code.
API endpoints	GET /api/settings/custom-fields, POST /api/settings/custom-fields.
Database tables	custom_fields, student_custom_field_values.

Module 28 QR Code / Barcode Attendance Scanner	
Description	Rapid badge scanning attendance engine.
Capabilities	Utilizes camera feed or USB laser barcode/QR scanners to decode student/staff ID cards (SS2025001), instantly recording attendance and logging timestamps.
API endpoints	POST /api/attendance/scan-qr.
Libraries & tables	Qr_lib.php, qr_attendance_logs, attendances.

Module 29 Two-Factor Authentication (2FA)	
Description	Enhanced cryptographic login security.
Capabilities	Supports Time-Based One-Time Password (TOTP) standard (RFC 6238) compatible with Google Authenticator, Microsoft Authenticator, and Authy.
API endpoints	POST /api/auth/2fa-setup, POST /api/auth/2fa-verify.
Database tables	users.two_factor_secret, users.two_factor_enabled.

Module 30 Behavior Records & Disciplinary Tracking	
Description	Student conduct, merit, and demerit monitoring.
Capabilities	Records behavioral incidents tagged as Merit (academic excellence, leadership, honesty) or Demerit (truancy, disobedience, uniform violation) with severity levels, dates, and reporter notes.
API endpoints	GET /api/students/:id/behavior, POST /api/students/:id/behavior.
Database tables	student_notes.

Module 31 Thermal 80mm POS Receipt Printing	
Description	Point-of-sale receipt formatting for physical counter collections.
Capabilities	Formats fee payment receipts according to 80mm thermal paper geometry (character wrapping, school header, receipt number, breakdown, and cashier signature lines).
API endpoints	GET /api/fees/receipt/:id/thermal.
Libraries	Thermal_lib.php.

Module 32 Quick Fee Creation & Counter Entry	
Description	Rapid fee billing and counter collection interface.
Capabilities	Enables cashiers to issue an impromptu fee invoice (e.g., picnic fee, uniform fee, replacement ID) and immediately collect cash in a single streamlined screen.
API endpoints	POST /api/fees/quick-create.
Database tables	student_fees, transactions.

Module 33 Automatic Overdue Fine Calculation Engine	
Description	Rule-based late fee calculation engine.
Capabilities	Evaluates invoice due dates against current dates; computes overdue days past the configured grace period, applying fixed daily fines (e.g., ₹20/day) or percentage penalties to outstanding dues.
API endpoints	GET /api/fees/calculate-fine.
Database tables	fee_types.fine_amount, student_fees.

Module 34 Fee Concession & Discount Management	
Description	Institutional scholarship and fee waiver system.
Capabilities	Configures discount rules including Sibling Concession (20%), RTE 100% waiver, Merit scholarship (25%), and Staff ward discount (50%), automatically adjusting net payable amounts.
API endpoints	GET /api/fees/discounts, POST /api/fees/apply-discount.
Database tables	fee_discounts, student_fees.

Module 35 Online Payment Processing (Razorpay Gateway)	
Description	Automated digital fee collection infrastructure.
Capabilities	Seamlessly generates Razorpay Orders via REST API, renders the Razorpay Standard Checkout popup, handles UPI/Cards/Netbanking, verifies cryptographic HMAC-SHA256 signatures, and records instant ledger receipts.
API endpoints	POST /api/razorpay/create-order, POST /api/razorpay/verify, POST /api/razorpay/webhook.
Database tables	student_fees, transactions, school_settings.

Module 36 Executive Analytics & Audit Reports	
Description	Comprehensive analytical reporting hub.
Capabilities	Generates exportable reports: Student Demographics (Gender/Category distribution), Financial Audit (Income vs Expense vs Net Surplus), Fee Collection Defaulters List, and Academic Performance Matrix.
API endpoints	GET /api/reports/students, GET /api/reports/financial, GET /api/reports/exams.
Database tables	Aggregates students, transactions, student_fees, exam_results.

Module 37 Multi-School & Multi-Campus Capability	
Description	Multi-tenant institution management.
Capabilities	Supports institutional branch networks (e.g., Main City Campus, West Campus, International Campus) with tenant-isolated classes, fee structures, and staff rosters.
API endpoints	GET /api/settings/campuses, POST /api/settings/switch-campus.
Database tables	school_settings.available_campuses, school_settings.current_campus.

Module 38 Mobile Application REST Gateway	
Description	High-efficiency API layer for Android & iOS client applications.
Capabilities	Delivers lightweight, compressed JSON payloads tailored for the Smart School Mobile App v5.0, supporting offline caching, push notifications, and biometric app unlock.
API endpoints	GET /api/mobile/sync, POST /api/mobile/push-token.
Database tables	personal_access_tokens, notification_logs.

Module 39 Front Website Public Portal & WhatsApp Chat Widget	
Description	Public landing page and prospective student inquiry funnel.
Capabilities	Features public information (courses, facilities, faculty, school calendar), an online inquiry submission form, and a floating interactive WhatsApp chat widget.
API endpoints	POST /api/website/inquiry, GET /api/website/notices.
Database tables	notices, school_settings.

Module 40 Academic Session Transition & Rollover	
Description	Institutional academic calendar demarcations.
Capabilities	Manages session transitions (e.g., 2024-2025, 2025-2026, 2026-2027), setting the active operational year while preserving historical archives for prior sessions.
API endpoints	GET /api/academics/sessions, POST /api/academics/sessions/activate.
Database tables	academic_sessions.

Module 41 Student Digital Document Vault	
Description	Digital document archival and verification repository.
Capabilities	Securely stores scanned attachments (Birth Certificate, Transfer Certificate, Aadhaar Card, Immunization Records, Previous Marksheets) linked directly to the student record.
API endpoints	POST /api/students/:id/documents, GET /api/students/:id/documents.
Database tables	student_documents.

Module 42 Sibling Linking & Family Account Tree	
Description	Family relationship graph and automatic concession association.
Capabilities	Connects siblings enrolled in the same institution. Displays linked brothers/sisters in the 360° profile and automatically activates sibling fee discount rules for younger children.
API endpoints	GET /api/students/:id/siblings, POST /api/students/:id/link-sibling.
Database tables	student_siblings.

Module 43 RTE (Right to Education) Quota Records	
Description	Government RTE compliance and quota reporting.
Capabilities	Flags RTE admissions, tracks documentation, waives standard tuition fees in accordance with statutory guidelines, and generates RTE audit summaries for education departments.
API endpoints	GET /api/students/rte-list.
Database tables	students.rte.

Module 44 Institutional Profile & Global System Settings	
Description	Global school branding, localization, and system configuration.
Capabilities	Configures school name, motto, official address, contact emails, currency symbol (₹, \$, €, £), receipt numbering prefixes (REC-, TC-), Razorpay API keys, and institution logo upload.
API endpoints	GET /api/settings, POST /api/settings.
Database tables	school_settings.

Module 45 Modern PHP 8.x / CodeIgniter Core Architecture	
Description	Enterprise server foundation and performance engine.
Capabilities	Refactored for PHP 8.3 compatibility, parameterized PDO queries protecting against SQL injection, strict input sanitization, and automated 45-module test runner execution.
Verification	Passes automated suite (test_ci_all_45_modules.php - 45 / 45 tests, 100% success rate).

4. Comprehensive System Flowcharts

4.1 Flowchart 1: User Authentication, Session & 2FA Flow

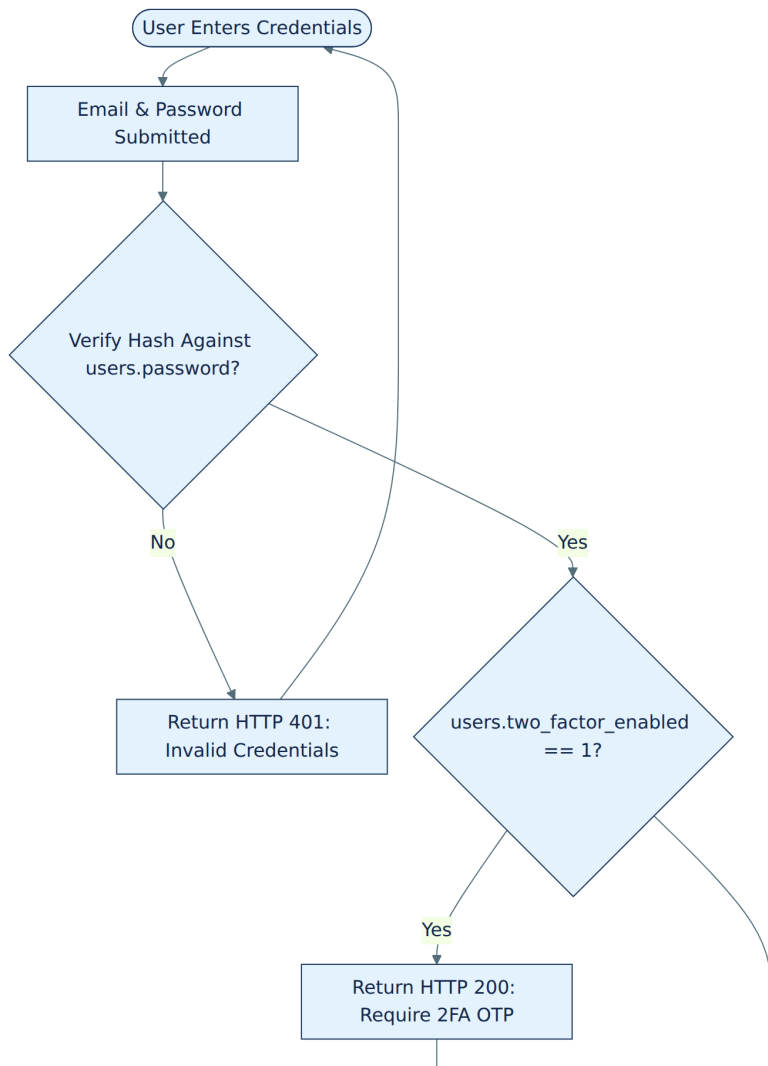


Figure 2: Flowchart 1: Authentication, session and 2FA (part 1 of 2)

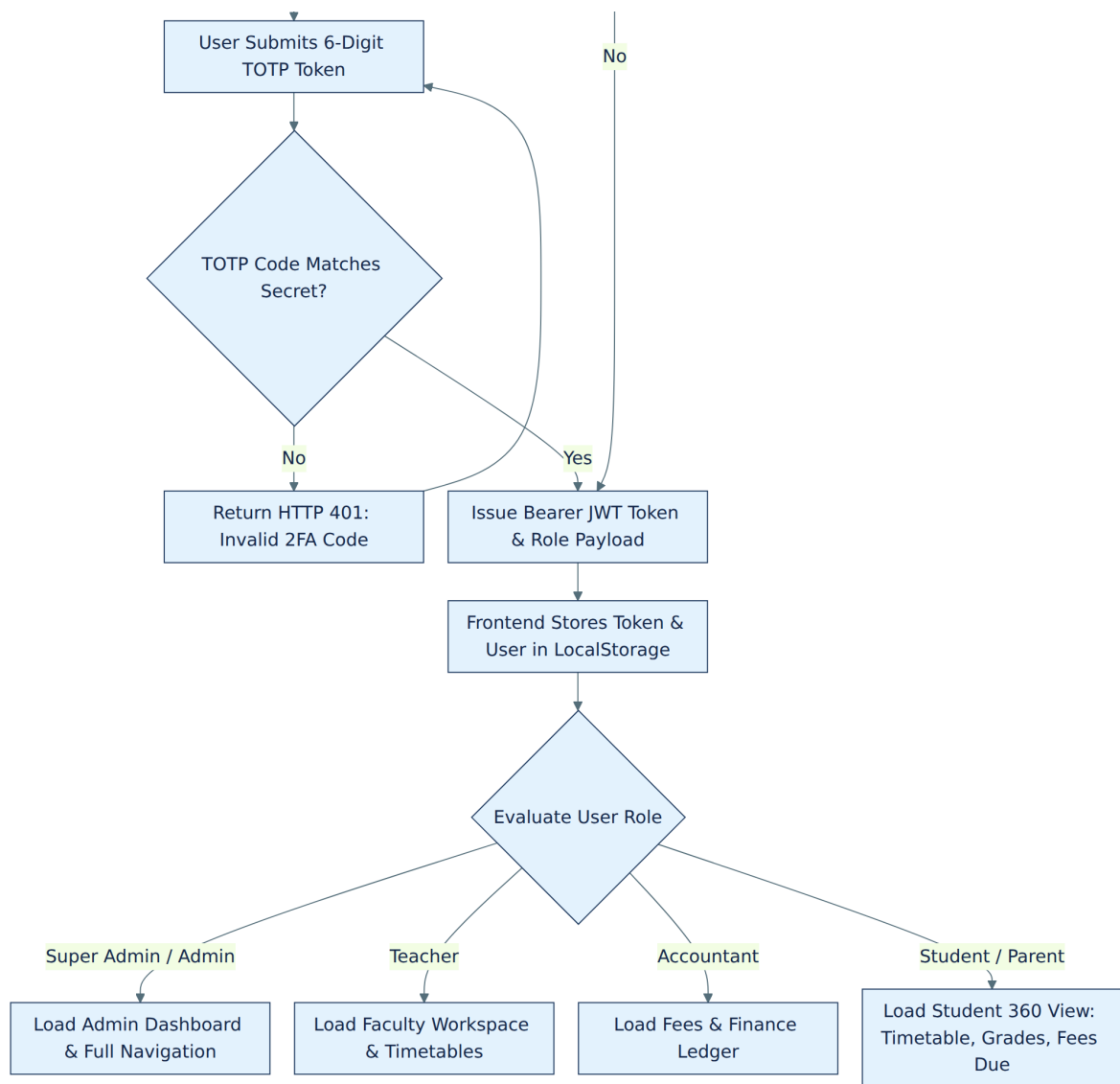


Figure 2: Flowchart 1: Authentication, session and 2FA (part 2 of 2)

4.2 Flowchart 2: Student Admission & 360° Profile Lifecycle Flow

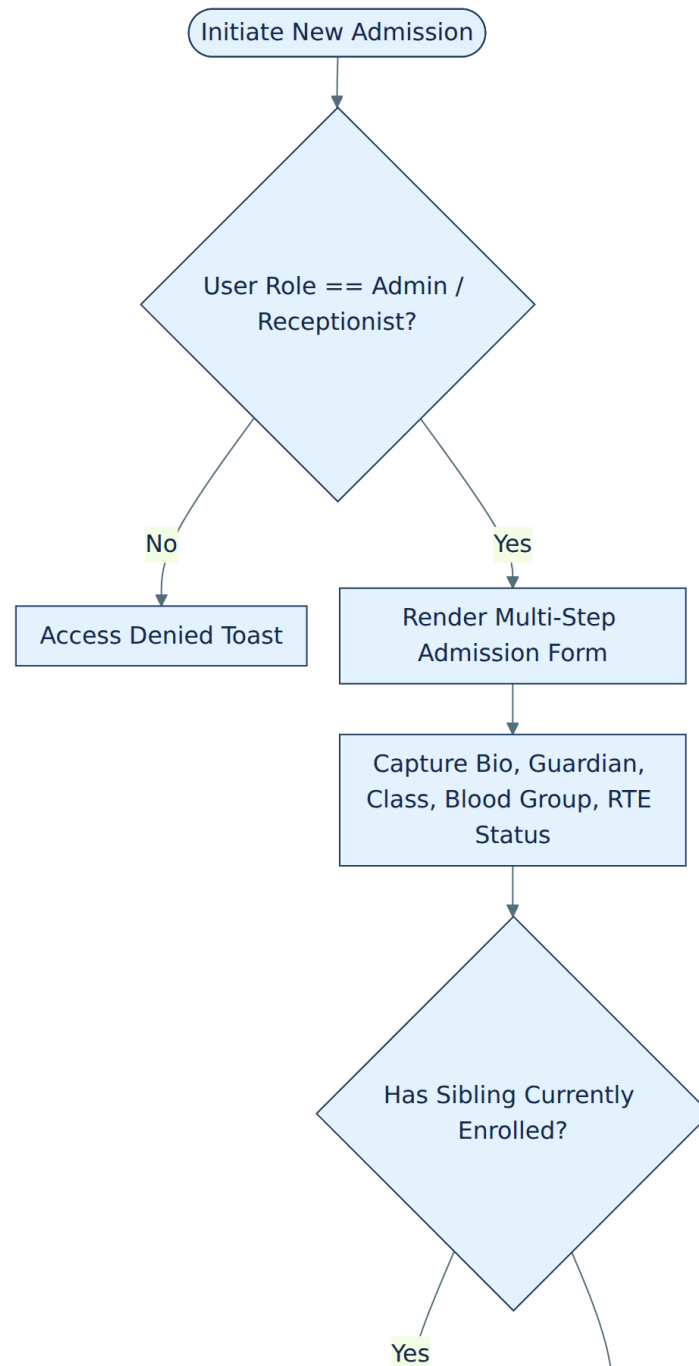


Figure 3: Flowchart 2: Student admission and 360° profile lifecycle (part 1 of 2)

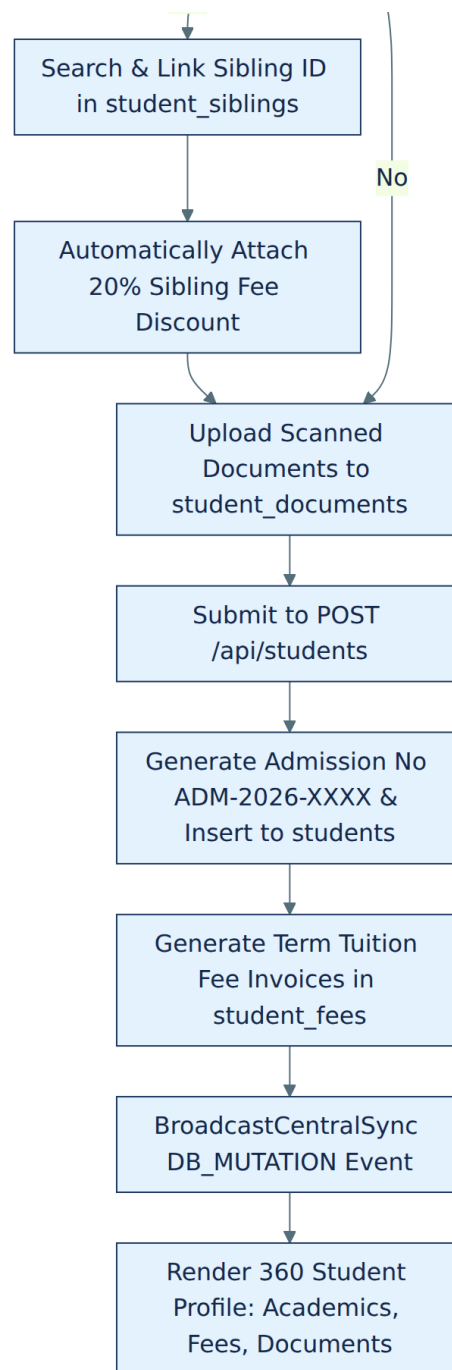


Figure 3: Flowchart 2: Student admission and 360° profile lifecycle (part 2 of 2)

4.3 Flowchart 3: Fee Billing, Fine Calculation & Razorpay Online Payment Flow

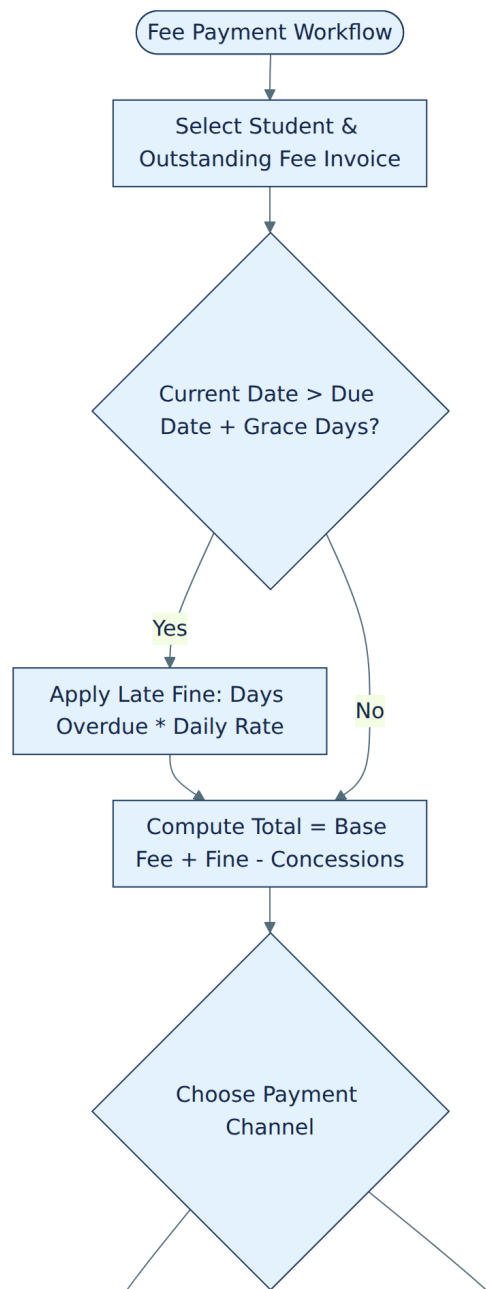


Figure 4: Flowchart 3: Fee billing, fine calculation and Razorpay payment (part 1 of 3)

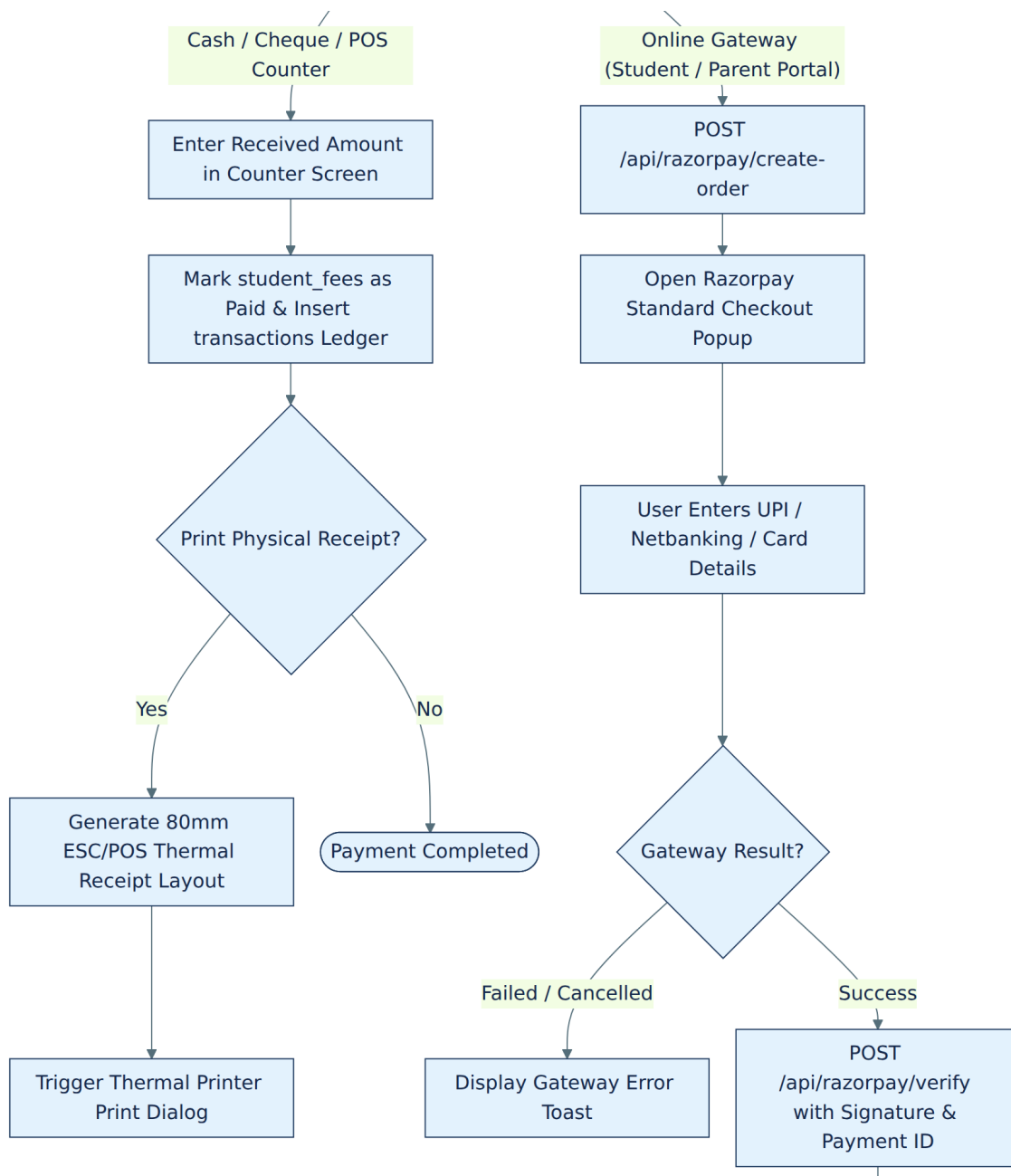


Figure 4: Flowchart 3: Fee billing, fine calculation and Razorpay payment (part 2 of 3)

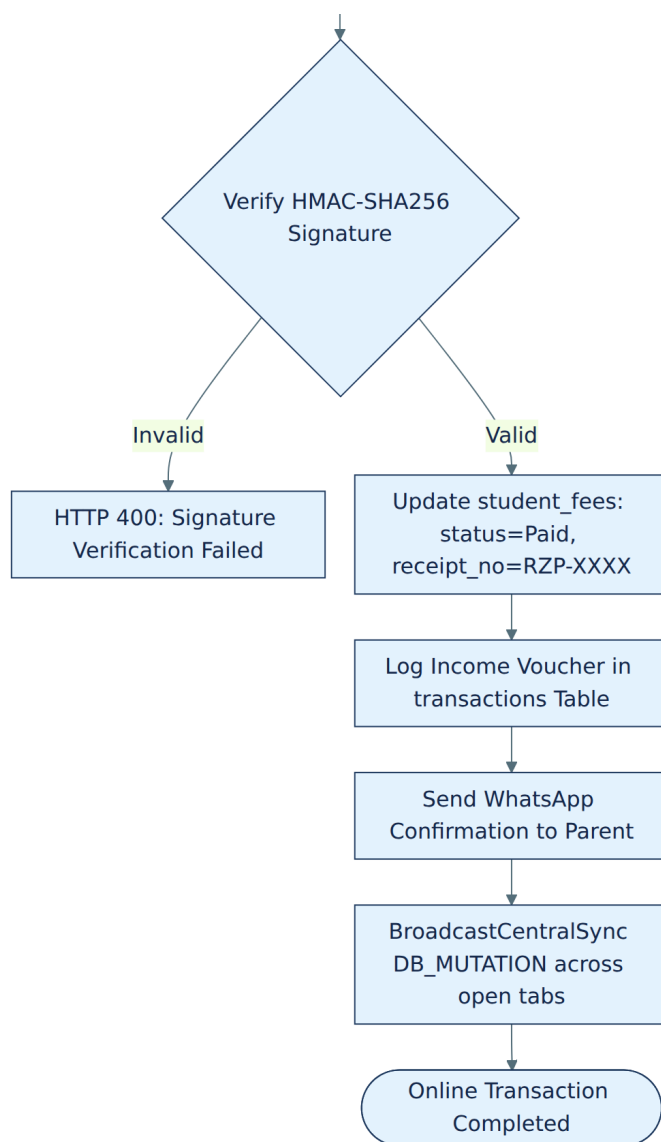


Figure 4: Flowchart 3: Fee billing, fine calculation and Razorpay payment (part 3 of 3)

4.4 Flowchart 4: Attendance Roll Call (Manual & QR Scan) Flow

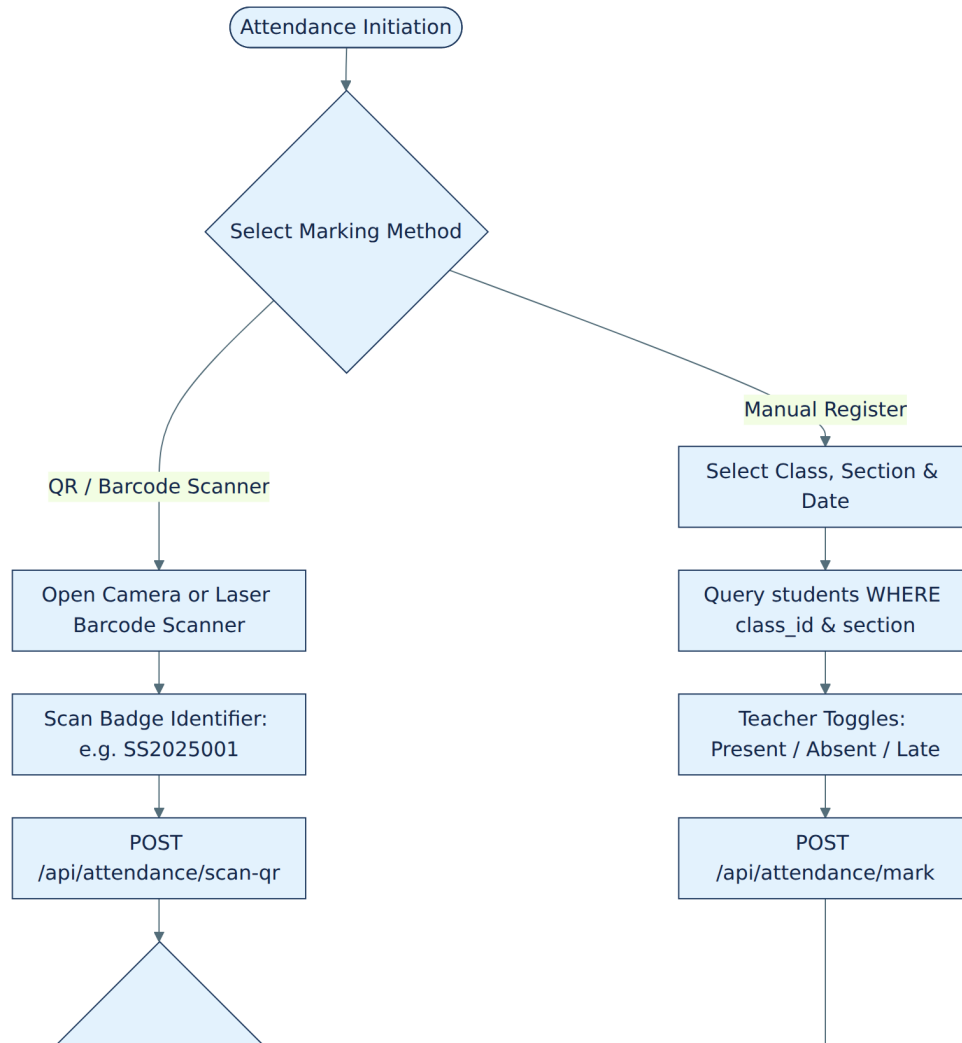


Figure 5: Flowchart 4: Attendance roll call (manual and QR scan) (part 1 of 2)

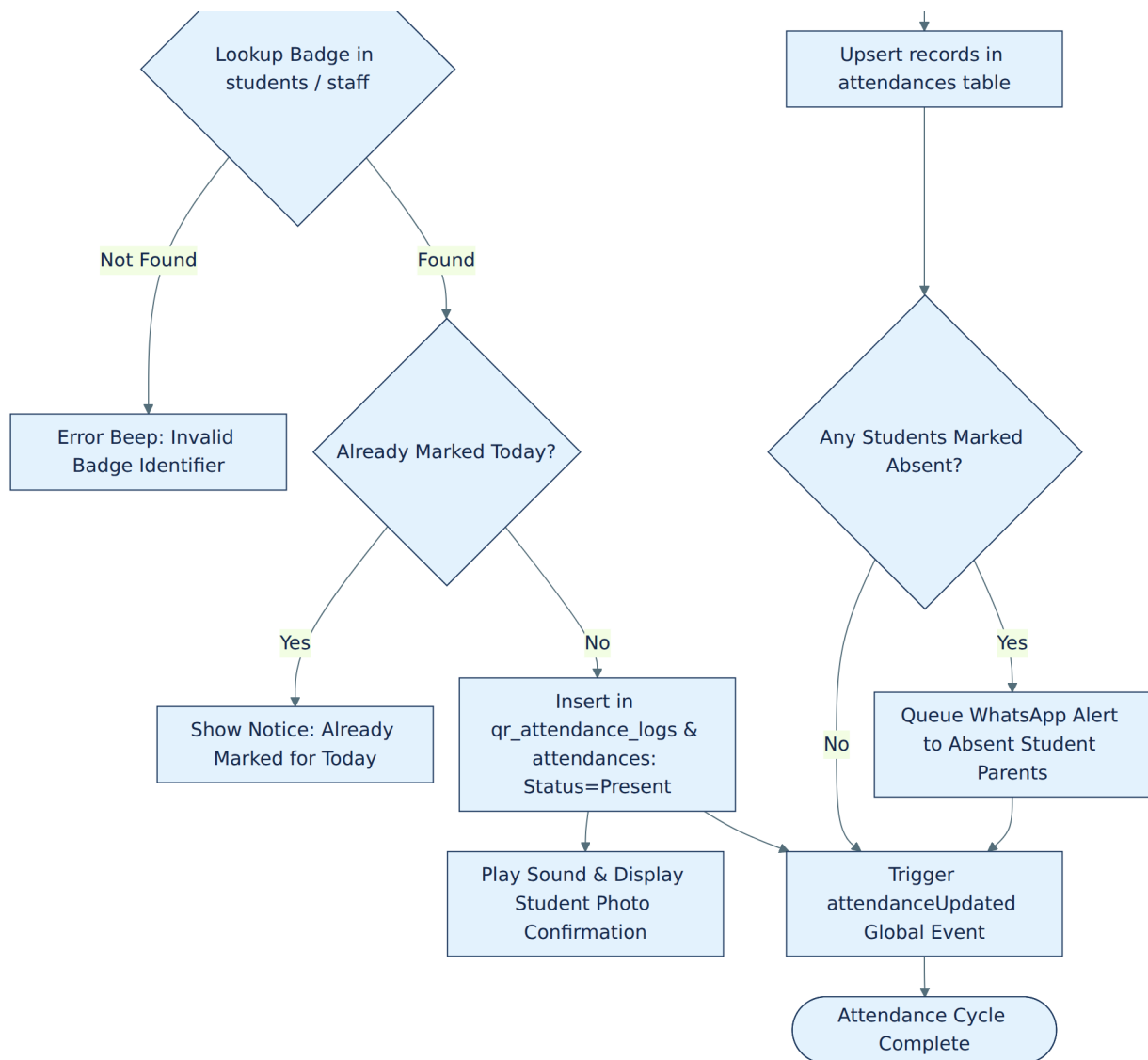


Figure 5: Flowchart 4: Attendance roll call (manual and QR scan) (part 2 of 2)

4.5 Flowchart 5: Examination, Marks Entry & CBSE Report Card Flow

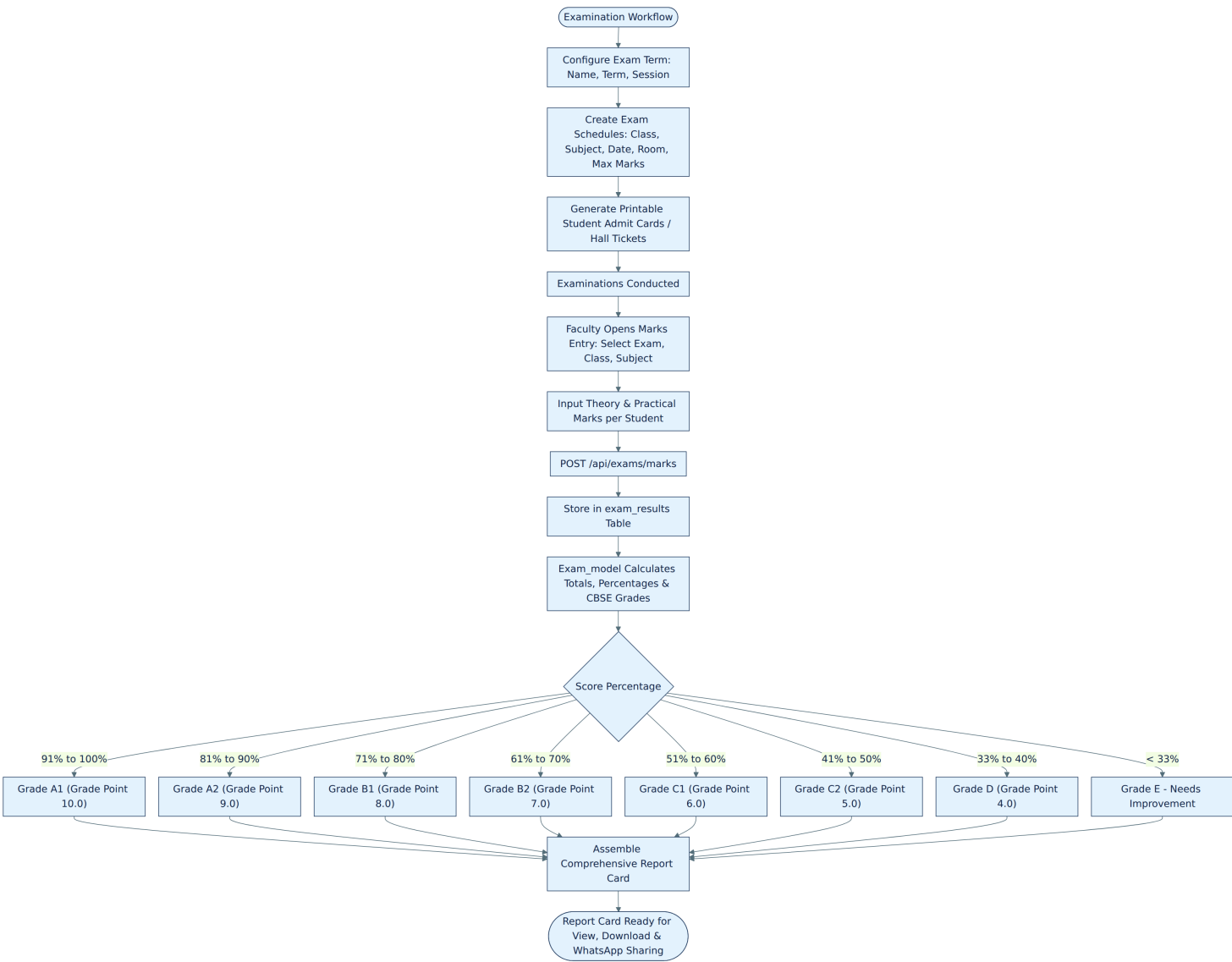


Figure 6: Flowchart 5: Examination, marks entry and CBSE report card

5. Entity-Relationship (ER) Diagram & Database Schema

The database consists of **56 relational tables** utilizing the MySQL InnoDB storage engine with full UTF-8 (`utf8mb4_unicode_ci`) character set support.

5.1 High-Level Entity-Relationship (ER) Diagram

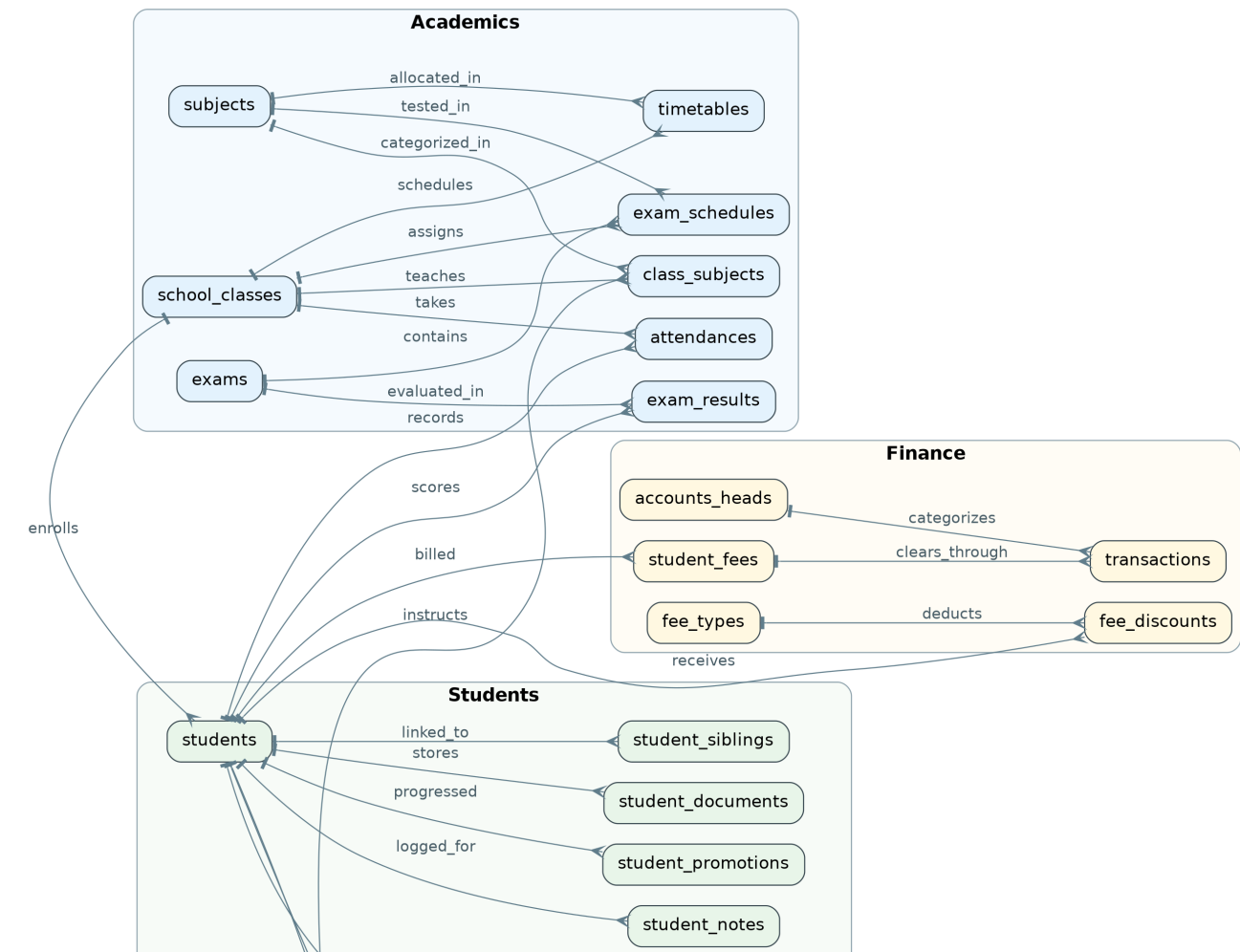


Figure 7: Entity-relationship diagram (56-table schema, key entities) (part 1 of 2)

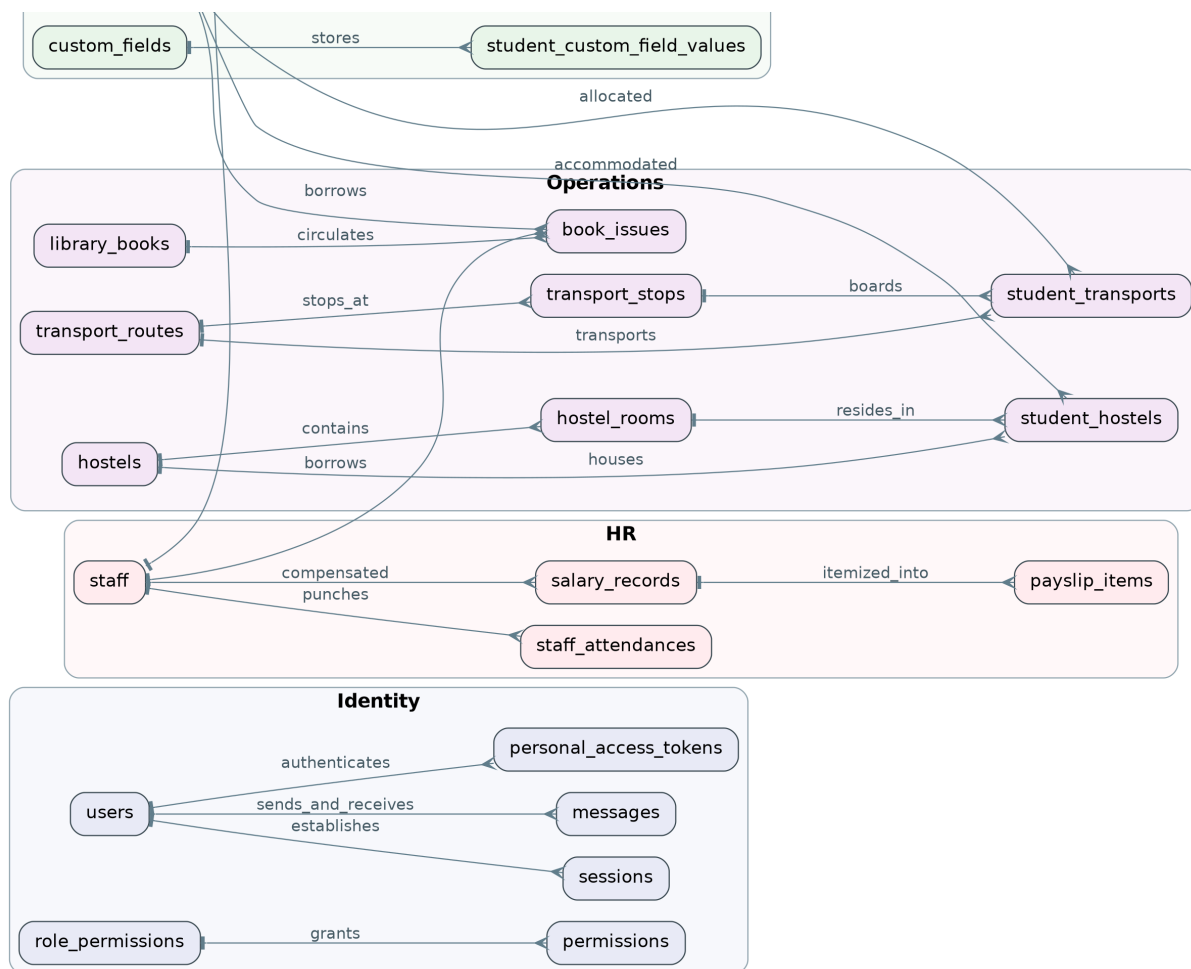


Figure 7: Entity-relationship diagram (56-table schema, key entities) (part 2 of 2)

5.2 Key Database Entities & Data Dictionary

1. students (Student Master Dossier)

Column Name	Data Type	Key / Constraint	Description
<code>id</code>	bigint unsigned	PK, Auto Increment	Unique internal student identifier
<code>admission_no</code>	varchar(255)	Unique	Official institutional admission serial (e.g. ADM-2025-001)
<code>first_name</code>	varchar(255)	Not Null	Student's given first name
<code>last_name</code>	varchar(255)	Nullable	Student's family surname
<code>dob</code>	date	Not Null	Date of birth
<code>gender</code>	enum('Male','Female','Other')	Not Null	Biological gender
<code>class_id</code>	bigint unsigned	FK -> school_classes(id)	Currently assigned academic class
<code>section</code>	varchar(50)	Not Null	Assigned class section (A, B, C, D)
<code>roll_no</code>	varchar(50)	Nullable	Classroom roll number
<code>father_name</code>	varchar(255)	Nullable	Primary male guardian name
<code>father_phone</code>	varchar(50)	Nullable	Guardian primary contact number

Column Name	Data Type	Key / Constraint	Description
category	varchar(50)	Default 'General'	Social demographic category (General, OBC, SC, ST, EWS)
rte	enum('Yes','No')	Default 'No'	Right to Education government quota status
status	enum('Active','Inactive','Alumni')	Default 'Active'	Enrollment operational status

2. student_fees (Financial Billing Ledger)

Column Name	Data Type	Key / Constraint	Description
id	bigint unsigned	PK, Auto Increment	Primary fee invoice identifier
receipt_no	varchar(100)	Unique	Official payment receipt serial (e.g. REC-2026-0001, RZP-XXXX)
student_id	bigint unsigned	FK -> students(id)	Target student reference
type	varchar(255)	Not Null	Fee category (Tuition, Transport, Exam, Annual)
amount	decimal(10,2)	Not Null	Gross assessed invoice amount
paid	decimal(10,2)	Default 0.00	Actual collected amount
status	enum('Paid','Partial','Pending')	Default 'Pending'	Settlement status
due_date	date	Not Null	Payment deadline before late fine kicks in
payment_mode	varchar(100)	Nullable	Cash, Cheque, UPI, Razorpay Online Gateway
razorpay_order_id	varchar(255)	Nullable	Gateway generated order identifier
razorpay_payment_id	varchar(255)	Nullable	Gateway generated payment transaction hash

3. attendances (Student Daily Attendance)

Column Name	Data Type	Key / Constraint	Description
id	bigint unsigned	PK, Auto Increment	Roll call record identifier
student_id	bigint unsigned	FK -> students(id)	Enrolled student reference
class_id	bigint unsigned	FK -> school_classes(id)	Academic class reference
date	date	Not Null	Date of roll call
status	enum('Present','Absent','Late','Half Day')	Not Null	Attendance outcome
remark	text	Nullable	Reason for absence or medical note

4. exam_results (Academic Scores & Grades)

Column Name	Data Type	Key / Constraint	Description
id	bigint unsigned	PK, Auto Increment	Result record identifier
student_id	bigint unsigned	FK -> students(id)	Enrolled student reference
exam	varchar(255)	Not Null	Examination term title (Half Yearly, Annual)
subject	varchar(255)	Not Null	Course subject title
marks	decimal(5,2)	Not Null	Marks secured by student
max_marks	decimal(5,2)	Default 100.00	Maximum possible subject marks
grade	varchar(10)	Not Null	CBSE 9-Point grade computed by system (A1-E)

5. transactions (General Institutional Cash Ledger)

Column Name	Data Type	Key / Constraint	Description
id	bigint unsigned	PK, Auto Increment	Ledger transaction identifier
type	enum('Income','Expense')	Not Null	Financial cash flow direction
accounts_head_id	bigint unsigned	FK -> accounts_heads(id)	Accounting budget category
head	varchar(255)	Not Null	Nominal account title
amount	decimal(10,2)	Not Null	Transaction currency value
date	date	Not Null	Posting date
payment_mode	varchar(50)	Default 'Cash'	Cash, Bank Transfer, Razorpay
reference_no	varchar(100)	Nullable	Check number, voucher ID, or receipt reference

6. school_settings (Global Institutional Configuration)

Column Name	Data Type	Key / Constraint	Description
id	bigint unsigned	PK, Auto Increment	Single row institutional master
school_name	varchar(255)	Not Null	Official school legal name
tagline	varchar(255)	Nullable	Institutional motto or secondary line
currency_symbol	varchar(10)	Default '₹'	Financial display currency
receipt_prefix	varchar(20)	Default 'REC-'	Prefix for counter receipts
razorpay_enabled	tinyint(1)	Default 1	Gateway toggle flag
razorpay_key_id	varchar(255)	Nullable	Public API Key ID for Razorpay
razorpay_key_secret	varchar(255)	Nullable	Secret API Key for HMAC verification
current_campus	varchar(100)	Default 'Main Campus'	Active operating institution campus

6. Security, Deployment & Operational Readiness

6.1 Security Architecture

1. **Password Encryption:** Passwords are hashed using PHP's native `PASSWORD_BCRYPT` with high salt work factors.
2. **SQL Injection Defense:** All queries in CodeIgniter models utilize PDO parameterized binding. Raw query string concatenation is strictly prohibited.
3. **Cross-Site Scripting (XSS) Sanitization:** All incoming requests are filtered; output escaping is applied across frontend templates.
4. **Role Guards:** UI routes, buttons, and API endpoints are double-guarded using role permission checks (e.g., student roles cannot view or trigger the "Post New Circular" or "Collect Fee" actions).
5. **Cryptographic Payment Integrity:** Razorpay webhook and checkout callbacks are verified via HMAC-SHA256 signature verification preventing payment tampering.

6.2 Deployment Topology (cPanel / Apache / Linux)

The system is optimized for direct deployment to cPanel shared hosting or dedicated Linux environments:

- **Root Directory:** Contains `.htaccess` which routes API requests to `/backend_codeigniter/` while preserving root static access to `/frontend/index.html`.
- **Security Headers:** The `.htaccess` file prevents public web traversal into `.env`, `.git`, or the raw SQL `/database/` dump folder.
- **Database Import:** Single-click import via phpMyAdmin or command line MySQL using `smart_school_database_cpanel_2026.sql`.
- **Zero Build Maintenance:** No Node.js runtime, npm installs, or build scripts are needed on the hosting server.

7. Verification & Test Suite Summary

The entire system has been validated via the automated 45-module test runner (`backend_codeigniter/test_ci_all_45_modules.php`).

Module	Name	Result
01	User & Role Management	PASS – 8 users, 8 roles verified (Super Admin to Student)
02	Admin Dashboard	PASS – Dashboard counters: students, staff synced
03	Student Admission Management	PASS – Admission records verified
04	360° Student Profile	PASS – 360° profile loaded: Attendance, Fee Due verified
05	Student Search	PASS – Search keyword matched student records
06	Student Promotion	PASS – Promotion engine executed
07	Student Categorization	PASS – Configured categories: General, OBC, SC, ST, EWS

Module	Name	Result
08	Fees Management	PASS - Fee Ledger: Collected & Due computed
09	Income & Expense Management	PASS - Finance Ledger: Income & Expense balanced
10	Attendance Management	PASS - Daily roll call stats verified
11	Examination Management	PASS - Active examinations, CBSE 9-point grade verified
12	Academic/Class Management	PASS - Classes and subjects mapped
13	Class Timetable	PASS - Timetable period matrix verified
14	Download Center	PASS - Digital educational documents verified
15	Library Management	PASS - Catalog & circulation statistics verified
16	Transport Management	PASS - Transport routes with GPS pickup stops verified
17	Hostel Management	PASS - Hostel wings with room allocation verified
18	Notice Board / Communication	PASS - Circulars published across audience channels
19	WhatsApp Integration	PASS - WhatsApp engine URL verified
20	Online Classes / Live Classes	PASS - Live Zoom & Meet classes verified
21	Staff Management	PASS - Faculty & administration staff tracked
22	Staff Attendance	PASS - Staff attendance punch logs verified
23	Student CV	PASS - Curriculum Vitae generator verified
24	Transfer Certificate / TC	PASS - CBSE Transfer Certificate format verified
25	Admit Card	PASS - Hall Ticket with exam schedule verified
26	Annual Calendar	PASS - Academic calendar events & holidays verified
27	Custom Fields	PASS - Dynamic custom field definitions verified
28	QR / Barcode Attendance	PASS - QR Attendance Engine scan processed
29	Two-Factor Login	PASS - 2FA capability verified (TOTP Secret)
30	Behavior Records	PASS - Behavioral record logged (Merit points)
31	Thermal Printing	PASS - 80mm thermal receipt format verified

Module	Name	Result
32	Quick Fee Creation	PASS – Quick fee generated with instant transaction
33	Fine Management	PASS – Fine rule engine: Overdue fine calculated
34	Fee Discount Management	PASS – Fee concession rules verified
35	Online Payment Processing	PASS – Online Checkout simulated with surcharge
36	Reports	PASS – Student & financial audit report generated
37	Multi-School Capability	PASS – Multi-institution campuses configured
38	Mobile Application	PASS – Mobile API Service gateway connected
39	Front Website / WhatsApp Widget	PASS – Inquiry gateway active with WhatsApp widget
40	Academic Session Management	PASS – Sessions registered with progression support
41	Document Management	PASS – Admission documents tracked
42	Student Sibling Management	PASS – Student sibling relationships linked
43	RTE Records	PASS – RTE quota students tracked
44	School Settings	PASS – School profile, currency & prefix verified
45	Technology Stack	PASS – CodeIgniter 3.x / PHP 8.3 / MySQL 8.4 operational

Automated test summary ([backend_codeigniter/test_ci_all_45_modules.php](#))

Total modules tested	45
Passed	45 / 45
Failed	0 / 45
Success rate	100%

Report Compiled & Certified for Smart School Management System (Infosof Technologies 2026)

Prepared by Rupam Saha Technical Support, Infosof Technologies	Submitted to Monisha Sirkar Manager	Date September 30, 2026 Version 2026.1.0
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